Brandon Yifan Yang

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Education

- **University of Pennsylvania** — Philadelphia, PA
  M.S.E. in Robotics — *Aug 2025 - May 2027*
- **University of Virginia** — Charlottesville, VA
  B.S. in Computer Science with Highest Distinction (3.9/4.0) — *August 2021 - May 2025*
- **Relevant Coursework**: Physical Intelligence*, Learning in Robotics*, GPU Programming & Architecture*, Machine Learning, Reinforcement Learning*, Natural Language Processing*, Probabilistic ML*, Optimization*
  *Graduate-level courses.

Experience

Research Lead — San Francisco, CA
AfterQuery — *May 2026 - Present*

- Leading a 6-person team on frontier AI-agent evaluation research, with a focus on long-context reasoning, multi-turn interaction, simulated users, and computer-use environments.
- Researching scalable robot data-collection methods across UMI-style teleoperation, egocentric recordings, and video datasets for training generalist robot policies.

Embodied Intelligence Research Intern — Beijing, China
Spirit AI — *May 2025 - Aug 2025*

- Trained and deployed Vision-Language-Action (VLA) model variants in PyTorch on a dual-arm robot, experimenting with contrastive objectives, observation-noise curricula, reduced guidance inputs, and mixture-of-experts (MoE).
- Designed and implemented DAgger data-collection pipeline for VLA models, including operator-takeover logic and post-training workflows for scaling corrective demonstrations.
- Integrated robotics simulation environments into a unified, reproducible setup, enabling efficient evaluation of VLA models; containerized training and evaluation workflows with Docker and Slurm. Open-sourced codebase and datasets on GitHub.

Research Experience

General Robotics, Automation, Sensing, and Perception Lab, UPenn — Philadelphia, PA
Advisor: Dinesh Jayaraman, Junyao Shi — *Aug 2025 - Present*

- Designing retrieval-augmented VLA models for robot manipulation.
- Building scalable data pipeline, training and evaluation infrastructure with Docker, Slurm, JAX, Ray, and Hugging Face on GPU clusters.

Learning and Interactive Robotics, University of Virginia — Charlottesville, VA
Advisor: Yen-Ling Kuo — *Aug 2024 - May 2025*

- **Interpretable Vision-Language-Action Models via Skill Conditioning**
  - Led *SkillVLA*, a skill-conditioned VLA model for language-conditioned manipulation with improved action interpretability via subgoal instructions and learned skill library.
  - Presented as an oral talk at the 2024 UVA LLM Workshop (slides), earning the Audience Choice Award (top 3 of 28 presentations).
- **Contrastive Learning for Robot Manipulation**
  - Performed contrastive learning over action sequences to learn behavior-grounded visual embeddings, improving visuomotor policies under heterogeneous camera poses and object appearances. (CoRL 2025)
  - Modified simulation environments and trained PyTorch-based visuomotor policies to evaluate learned embeddings.

Research Assistant, Collaborative Robotics Lab, University of Virginia Charlottesville, VA
Advisor: Tariq Iqbal *May 2022 - May 2024*

- **Grounded Location for Object Manipulation (GLOMA)** [code]
  - Led team of 3 to develop zero-shot image-editing model grounded by language instructions for object relocation and manipulation tasks, designed for downstream robotic applications using goal-conditioned RL and Behavioral Cloning (BC).
  - Integrated language grounding with visual perception by using bounding box guidance from pre-trained language models, enabling precise object relocation without external supervision and improving baseline performance by 65%.
  - Collected and annotated custom dataset for fine-tuning pre-trained language and vision models.
  - Presented poster at 3 conferences and symposiums.
- **Centralized multi-agent RL for Collaborative Tasks** [code]
  - Developed long-horizon on/offline centralized MARL for robotic bolt screwing tasks.
  - Designed and optimized custom reward functions in multi-agent framework for task completion and agent collaboration, improving task success rate by 20%.
  - Deployed and tested custom simulated environments in IsaacGym for training and evaluation.

Publications

Jiani Huang*, **Brandon Y. Yang***, Amish Sethi, Yuchen Zheng, Christopher Watson, Jianing Qian, Mayur Naik, and Dinesh Jayaraman. "Retrieval-Augmented Vision-Language-Action Model." Submitted to *CoRL 2026*.

Amish Sethi*, **Brandon Y. Yang***, Michael Evans, Spencer Mateega, and Sam Jung. "Position: Trust No Multi-Turn Benchmark Without Measuring Its User Simulator." Submitted to *NeurIPS 2026 Position Paper Track*.

Lee, Sung-Wook, Xuhui Kang, **Brandon Y. Yang**, and Yen-Ling Kuo. "Class: Contrastive learning via action sequence supervision for robot manipulation." *Conference on Robot Learning*, pp. 4743-4766. PMLR, 2025. [Website][arXiv][PDF]

*Equal contribution.

Software Projects

openpi-cuda [GitHub]: Developed custom CUDA kernels via C++/Python bindings to accelerate $\pi_{0.5}$ VLA model inference; achieved 18.5ms latency reduction over baseline PyTorch.

notie-markdown [Website]: Markdown rendering web app with support for equation previewing, graphing, code running, and more. Built with React, TypeScript, Python, Flask. Deployed on Vercel and Heroku.

Blogs and Notes [blog, notes]: Detailed blog posts and course notes with visualizations, graphs, and code on CS and AI topics, written with notie-markdown.

SmartOH [GitHub]: Queue management system designed for office hours. Features real-time queue updates, notifications, and analytics. Built with Next.js, TypeScript, Python, FastAPI. Deployed with CI/CD pipeline on Vercel and AWS EC2.

Voy: Collaborated with 7 non-profits to develop Voy, a volunteer and driver management platform; competed and received $1000 in funding from UVA's Entrepreneurship Cup. Built with React, Node.js, TypeScript, JavaScript.

Skills

Programming Languages: Python, C/C++, Java, JavaScript, TypeScript

Tools & Frameworks: `uv`, `ruff`, `ty`, Claude Code, Codex, PyTorch, Jax, CUDA, Slurm, Ray

Last Updated: June 7, 2026